The Career Development Quarterly

WILEY

ARTICLE

A Case Study of High School Graduates' Reflections on Career Education: Insights From High-Stakes Context

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Received: 10 November 2025 | **Revised:** 5 June 2026 | **Accepted:** 19 June 2026

Keywords: career education program | context, input, process, and product (CIPP) | high school graduates | learning experience | learning outcomes

ABSTRACT

Career education in secondary schools has received increasing attention, yet its practices in high-stakes examination contexts (such as the Chinese college entrance examination) remain underexplored. This study addressed this gap by investigating graduates' reflections on a typical career education program at a large high school in Mainland China using the CIPP (context, input, process, and product) model. Data were collected through in-depth interviews with 15 graduates of varying academic achievement levels, who had participated in the program during its first 3 years of implementation. Thematic analysis uncovered the complex dynamics of program implementation and the program's role in students' personal development and career navigation during a critical and stressful period. The findings have implications for enhancing the effectiveness of career education, suggesting that more proactive approaches can be taken in high-stakes contexts, particularly through innovations in program planning, teacher expertise, and parental involvement.

1 | Introduction

To prepare youth for an era of increasing uncertainty, secondary-level career education has garnered significant global attention (Maree 2019; Plank et al. 2008). As a defining challenge in high-stakes educational contexts, the tension between intensive exam preparation (long daily hours in schools, leaving minimal time for rest or extracurricular activities) and holistic student development raises critical questions about the contextual effectiveness of career education (Chiang 2018). Although research supports the role of career education in enhancing students' career planning abilities (e.g., Gu et al. 2020; Røise 2022), the specific implementation challenges and the mechanisms of effectiveness within high-stakes examination systems remain underexplored. These gaps underscore the importance of examining students' lived experiences, particularly through the reflections of high school graduates, whose growing agency offers a unique and invaluable perspective on program effectiveness.

This research aims to address this gap by exploring a school-based career education program in a large, well-established public high school in Mainland China from the graduates' perspective. Mainland China offers a quintessential context for examining this challenge. The national college entrance examination (the "Gaokao") profoundly influences the entire secondary education system (Cheng and Hamid 2025). Notably, the reformed examination system, introduced in 2014, has significantly increased stakeholders' awareness of career planning, while also highlighting the challenges of balancing personal growth with examination success in school settings (Luo and Cao 2024). Therefore, guided by the CIPP model (context, input, process, and product; Stufflebeam 2003), this study examined the design and implementation of a career education program as perceived by graduates, as well as its impact on their learning experiences and outcomes.

1.1 | Washback Effects of the Reformed College Entrance Examination

Since 2014, Mainland China has gradually implemented a reformed national college entrance examination, granting students greater academic autonomy (Ling et al. 2024). The new system replaced the previous arts and science tracks with a more flexible model, allowing students to choose elective examination subjects alongside compulsory ones to determine their eligible university majors. Despite this flexibility, the system's high-stakes nature persists, as university slots are still allocated to the highest scoring applicants within each major.

Research indicates that the negative washback (i.e., how the intense pressure and focus of the college entrance examination shape and restrict daily teaching and learning activities) on stakeholders remains remarkably similar to the pre-reform era (Cheng and Hamid 2025; Yu and Su 2024). Students often prioritize exam-oriented learning, sacrificing intrinsic motivation and deep cognitive engagement (Yu et al. 2018). Parents who view university admission as a gateway to social status tend to be highly involved and set high expectations (Cheng and Hamid 2025). In turn, teachers often strategically align their pedagogy with exam requirements, sometimes at the expense of depth of knowledge (Chiang 2018; Meng et al. 2021).

The most significant shift brought by the reform appears to be the increased attention to career education. Previously neglected due to a focus on test-driven activities (Fan and Leong 2016), career education has been revitalized by the new system's requirement of linking selected examination subjects with university majors. This has boosted students' and parents' extrinsic motivation for academic and vocational planning and drawn administrative attention to school-based programs (Li and Xue 2022; Luo and Cao 2024). However, a fundamental tension exists between the high-stakes logic of the examination system, which emphasizes efficiency and performance, and the core of career education, which embraces personal experience and uncertainty (Chiang 2018; Healy 2023).

Beyond this underlying tension, the system imposes structural constraints on students' career planning. The reformed examination continues to require students to select majors before college admission, a process that often leads to immature choices made without sufficient exploration (Cheng and Hamid 2025). Furthermore, competitive university admissions policies can require students to enroll in undesired majors as a condition of admission. This forces students to make a difficult choice: invest years in a misaligned field or attend a less-prestigious institution (Heger 2017; Cheng and Hamid 2025). With very limited opportunities to switch majors later, students' academic and vocational trajectories can be severely disrupted by decisions made under this pressure. In summary, although the reformed examination system has drawn increased attention to career education, it has also imposed structural constraints on students' academic decisions due to its high-stakes nature.

1.2 | Effective Factors of Career Education

Research across various contexts has identified key factors that enhance the effectiveness of career education programs from students' and graduates' perspectives. Studies have shown that participatory and experiential design, as well as collaboration with significant others, may strengthen students' career awareness and practical planning skills and support subsequent career decisions (de Vries et al. 2024; Silipo and Caldon-Ruggles 2021; Yoel and Dori 2022). Moreover, effective teacher guidance and students' full engagement in career education both contributed to its effectiveness (de Vries et al. 2024; Goins 2015).

However, the applicability of these findings to high-stakes contexts is questionable, as they were primarily observed in systems with flexible college admission policies and abundant educational resources. Notably, in high-stakes environments like China's, the intense focus on exam preparation may monopolize school resources and students' time (Yu and Su 2024), leaving little capacity for the individualized and collaborative exploration deemed essential to effective career education. Gu et al. (2020) conducted a 5-week experimental career program for first-year high school students. Paradoxically, although the program increased students' career decision-making self-efficacy, it also led to greater perceived difficulties in career preparedness and in gathering information. The increased difficulties may reflect students' perceptions of the complexity of their future choices, leading to temporary self-doubt. However, few studies have examined the factors that led to these perceptions and self-doubt, or the long-term effects of such practices. Therefore, alternative effective factors in career education in high-stakes contexts deserve further exploration.

1.3 | CIPP Model in Program Evaluation

This study adopts the CIPP model (Stufflebeam 2003) as its guiding framework to systematically investigate students' perceptions of a school-based career education program within a high-stakes context. The CIPP model is designed to assess the conditions and implementation activities, together with outcomes of the program through four aspects: context (the needs, problems, and assets of the program's environment), input (the program's goals, resources, and strategy), process (the actual implementation, including planned and unplanned events), and product (the program's ultimate outcomes and impacts).

The CIPP model has been widely applied to evaluate the coherence and interrelations among program components across diverse educational settings. For instance, Vian (2023) used the CIPP framework to examine a translation course, finding a misalignment between the course's input and students' needs that negatively impacted its process and outcomes. Similarly, Rooholamini et al. (2017) evaluated a medical school's curriculum, confirming its alignment with ideal standards, whereas Misra and Dimri (2022) used the model to analyze learner feedback in a technology course, revealing that strong contextual support led to satisfactory outcomes. Collectively, these studies demonstrate the model's utility in diagnosing program strengths and weaknesses by considering the entire program ecosystem.

Given the identified tension between the ideals of career education and the realities of a high-stakes environment, the CIPP model is particularly well-suited for this study. It provides a structured lens for systematically analyzing graduates' reflections, allowing an examination of the interplay between the program's intended design (context, input) and its lived implementation (process), and how this dynamic shapes the final outcomes (product). Therefore, this study sought to answer the question: How do graduates perceive and reflect on their high school career education program in terms of its CIPP?

2 | Methods

Drawing on the retrospective perceptions of the first three graduating cohorts, this single-case study (Yin 2018) aimed at an in-depth investigation of the initial development and implementation of a school-based career education program within China's high-stakes educational environment.

2.1 | Selection of the Case

The case school was selected for its typicality, representing a large-scale, college-entrance-focused public high school that exemplifies the prevalent practice of career education in Mainland China (Pang 2019; Wang and Yang 2023). Located in an Eastern province, this prestigious local high school has approximately 1000 students per grade, with over 80% of its graduates admitted to undergraduate programs each year. Given the immense expectations from students, parents, and the government, this school provided a rich context to investigate the challenges of implementing career education in a high-stakes academic environment.

In response to the national college entrance examination reform, the school formally launched its career education program in 2021. The program was designed to cover three core areas: self-awareness, exploration of vocations and university majors, and management of the career decision-making process. The program was primarily delivered through large-group lectures, supplemented by class meetings. Given the lack of qualified full-time career education staff, program activities were managed by existing personnel (a senior administrator and class teachers). These practitioners collaboratively developed the program materials, drawing upon their personal and professional experiences. To confirm the case school's suitability, the first author contacted three teachers and one administrator through connections to conduct preliminary screening interviews and to review the program plans and teaching schedules.

2.2 | Participants

The participants included 15 graduates (ages 18–22; see Table 1 for demographic details) who attended the career education program during its first 3 years of implementation and were enrolled in their first to third year of higher education at the time the research was conducted. Participants were recruited and selected using a combination of snowball and purposive sampling strategies to capture the diversity of perspectives. In addition to graduation year and gender, which are known to influence perceptions of career programs, participants' academic achievements were also included as a stratified inclusion criterion. Given the high-stakes nature of the Gaokao, students' achievement levels may influence their academic decision-making and engagement with career education. Specifically, academic achievement was classified according to the selectivity of the universities attended, following China's national tiered system ("Double First-Class" institutions, regular undergraduate universities, and vocational colleges), a categorization that corresponds to their Gaokao scores.

Initial recruitment began through the first researcher's social connections, with subsequent participants referred by earlier ones. Throughout this process, sampling was deliberately controlled to ensure an approximate balance across graduation years (targeting five participants per cohort) and full representation of all academic achievement categories. Although gender balance was also a sampling aim, the final sample consisted of 10 young men and 5 young women, reflecting recruitment constraints and the limited accessibility of eligible female graduates within the researcher's network.

2.3 | Author Reflexivity

To maintain analytical sensitivity, reflexive practices were embedded throughout the research process. Although none of the authors had firsthand experience with the reformed Gaokao or the career education curriculum under the new system, all were educated within the Chinese secondary education context. This shared background provided familiarity with the high-pressure nature of high-stakes examinations and sensitized the team to participants' accounts of balancing personal interests with academic competitiveness.

At the same time, this positionality may have introduced the authors' experiences and values into the coding process. In particular, the first and second authors had personal experience with constrained time for career exploration in high school, as well as expectations that career education programs should provide sustained support for students' long-term career development. Without sufficient reflexive awareness, such assumptions could have led to an overemphasis on participants' frustrations and negative learning experiences during coding. These potential biases were identified through discussions among team members and were subsequently addressed through iterative dialogue, recoding, and ongoing reflexive examination, thereby enhancing the credibility and balance of the findings.

2.4 | Interview Protocol

This study received ethical approval from the research committee of City University of Macau, and written informed consent was obtained from all participants before data collection. Data were collected through semi-structured interviews conducted by the first author between September 2023 and January 2024. Each interview lasted approximately 1 h and was guided by an interview protocol structured around the four dimensions of the CIPP model (Stufflebeam 2003): context (e.g., participants' interests in career education), input (e.g., their understanding of the program's purpose and implementation), process (e.g., their learning experiences and related opinions), and product (e.g., how they were influenced by the program). Five interviews

TABLE 1 | Participants' background information.

No.	Gender	Age	Grade	Major	Academic achievement	Interview
1	Male	19	Freshman	Computer science and technology	High	Online
2	Female	19	Freshman	Biological engineering	Medium	Face-to-face
3	Female	18	Freshman	Dance performance	Low	Online
4	Male	18	Freshman	Clinical medicine	High	Face-to-face
5	Female	18	Freshman	Financial management	Medium	Online
6	Male	19	Sophomore	Computer Science and Technology	High	Face-to-face
7	Male	20	Sophomore	E-commerce	Medium	Online
8	Male	20	Sophomore	Traditional Chinese medicine	Low	Face-to-face
9	Male	20	Sophomore	Economics	High	Face-to-face
10	Female	20	Sophomore	Marketing	Medium	Face-to-face
11	Male	22	Junior	Electronic information science and technology	High	Online
12	Female	21	Junior	Architectural engineering technology	Medium	Face-to-face
13	Male	21	Junior	Cross-border E-commerce	Low	Face-to-face
14	Male	22	Junior	Integrated circuit	Medium	Face-to-face
15	Male	22	Junior	Cultural Industry Management	Medium	Face-to-face

were conducted online to accommodate participants' locations, and the remaining interviews were conducted face-to-face. All interviews were audio-recorded, transcribed, and uploaded to NVivo 14 software for further analysis.

2.5 | Analysis

Data analysis followed a hybrid inductive–deductive thematic analysis approach (Fereday and Muir-Cochrane 2006). Initially, the first author repeatedly read the transcripts to grasp essential meanings and identify meaningful units. Inductive codes were then generated directly from the data, from which subthemes emerged. These inductive subthemes were subsequently linked to the deductive codes corresponding to each CIPP dimension to address the research questions. For example, the inductive subtheme of "student needs" was connected to the deductive theme of "context," as it reflected participants' perceptions of contextual factors influencing their career education experiences.

To ensure the trustworthiness of the analysis and minimize potential researcher bias, we employed a rigorous, iterative coding process that included researcher triangulation. The second author independently coded all transcripts following the same steps. On the basis of cross-checking the coding performed by the first author, adjustments were proposed. The two authors then engaged in in-depth discussions to reconcile interpretations and reach a consensus on a preliminary coding scheme. This scheme was subsequently reviewed by the third and fourth authors to be further refined into the final coding system (Lincoln et al. 1985), as presented in Table 2. Furthermore, following the independent coding phase, the first and second authors engaged in continuous critical dialogues regarding the nuances of the interview data. These sustained discussions ensured that interpretations remained firmly grounded in the participants' experiences, thereby enhancing the overall rigor of the analysis.

3 | Findings

The findings reveal a complex dynamic regarding the high school career education program's role in empowering graduates' initial decision-making under intense academic pressures and systemic constraints. The findings are structured around the four dimensions of the CIPP model as main themes with subthemes detailed. Specifically, the theme context consists of the subthemes *school context* and *student needs*; input presents the program's *dual objectives* and *limited resources*; process details *course delivery* and

TABLE 2 | Categories, themes, and subcategories.

Themes	Subthemes	Codes
Context	School context	Supportive learning atmosphere
		Pressure to perform
	Student needs	Instrumental needs
		Curiosity
Input	Dual objectives	Promoting holistic development
		Facilitating examination preparation
	Limited resources	Program scheduling
		Staffing
		Learning materials
Process	Program delivery	Targeted content
		Direct instruction
		Impersonal support
	Student engagement	Emotional engagement
		Behavioral engagement
		Cognitive engagement
Product	Empowerment	Expanded options
		Self-understanding
		Informed decision-making
		Sense of direction
	Limited adaptability	Systemic constraints
		Family involvement

student engagement; and finally, product assesses the resultant *empowerment* alongside the program's *limited adaptability* to external constraints.

3.1 | Context

Participants perceived that although their schools provided material and teaching resources to support their studies, the general atmosphere exposed them to exam preparation pressure. In such an environment, participants at various levels needed the career education program for both its instrumental value in choosing examination subjects and its support for personal growth.

3.1.1 | School Context

Fourteen participants acknowledged that the school provided *a supportive learning environment*, emphasizing helpful peers, dedicated teachers, and adequate facilities. Participant 3 noted, "Our class atmosphere was good, and the school's multimedia equipment was also newly replaced." Participant 6 highlighted that "The teachers were excellent."

However, five participants noted that this supportive environment was overwhelmingly focused on exam preparation, intensifying *the pressure to perform*. As Participant 13 described, "The psychological environment in the school…lacks a balance between work and rest." This stemmed from the school's emphasis on college admission rates, pushing students to devote themselves to exams. Participant 14 observed, "In this environment…there is pressure, but also motivation."

In sum, participants perceived the school environment as supportive in terms of resources and teacher quality, yet highly exam-oriented, thereby simultaneously fostering motivation and intensifying academic pressure.

3.1.2 | Student Needs

In the aforementioned exam-oriented context, participants had limited opportunities to access career education, with none having prior exposure before high school. Their needs primarily stemmed from examination requirements, coupled with curiosity related to their developmental stage.

Fourteen participants expressed a prominent instrumental need for guidance on choosing examination subjects and future college majors. Specifically, they needed to clarify their academic–vocational trajectory:

> I don't know how to make choices or what to do, maybe it's because I don't have a direction for the future… (I hope) to accurately find that specific point needed for the future. (Participant 7)

Similarly, Participant 11 stated, "It is very hard to learn about specific majors in high school, as well as... what future jobs actually entail, and how they relate to personal interests. I feel it was really necessary."

Beyond examination requirements, five students harbored a curiosity about career planning, recognizing that both vocational and self-exploration are critical in late adolescence. Participant 9 commented, "I felt I needed to improve my understanding at the time and acquire more extensive knowledge." Participant 14 added, "I'm quite interested because it includes the next steps of my development. I'm also quite interested in personal development..." However, this curiosity had not yet encompassed the uncertainties and challenges involved in career development.

Overall, participants' needs were primarily driven by instrumental demands related to examination choices, whereas their emerging curiosity remained limited in depth and awareness of long-term uncertainties.

3.2 | Input

Participants generally perceived that the program aimed to promote their holistic development and their exam competitiveness. Despite acknowledging the program's importance, they also felt frustrated by the limited resources allocated.

3.2.1 | Dual Objectives

All participants emphasized the school's intention to connect the immediate challenge of selecting examination subjects with *promoting holistic development*. As Participant 10 indicated: "It is to make me think about my future and plan for my career or goals." Participant 2 further illustrated this:

> The significance is to make me willing to figure out which subject I should choose to help me find a job in the future, or to better enable me to realize my dreams and become the person I want to be.

Five students further associated the program's objective with *facilitating examination preparation* and the school's concern for college admission rates. Notably, these perceptions varied by academic performance. Higher achieving participants (P4, P6, and P11) viewed the program as beneficial for excelling in examinations through informed subject choices. As Participant 4 noted, "A part of it is also for our grades," and Participant 6 added, "The purpose is still to let us choose the subjects we are good at." Participant 11 further explained that beyond subject selection flexibility, "There is a partial requirement for the college admission rate." In contrast, lower achieving students (P3 and P13) perceived career education as a means of reinforcing the importance of academic effort. Participant 3 stated, "It urges students to study seriously and then find a good job," and Participant 13 similarly commented, "It is mainly about... encouraging us to study."

In sum, participants perceived the program's objectives in holistic development and examination preparation, reflecting their understanding of the school as both a career education provider and the gatekeeper of the high-stakes examination.

3.2.2 | Limited Resources

Participants frequently reported that the school invested very limited resources in the career education program compared to exam-preparation subjects, fearing this would have negative consequences.

Program scheduling sparked considerable discussion. Participants suggested a more systematic program schedule design to help students fully internalize the content. As Participant 1 noted: "If the classes were offered more frequently, such as once every two weeks, we would probably achieve better results over the course of a year."

Although the ideal starting timing varied among participants, seven participants agreed that the program should run throughout their entire high school. Participant 3 pointed out, "(Starting in) 12th grade is a bit too late... If they had told us these things in 10th or 11th grade, I might have thought about things differently." Participant 7 added, "I feel like we really need it in 11th and 12th grade as well."

While recognizing the program's value, participants still expressed concerns about potential conflicts with coursework. Participant 2 stated, "I think it could be done during extracurricular time, winter and summer vacations, or weekends, because we really have so much homework (during semesters)."

Staffing was also a key concern. Participants noted a significant disparity between the program's scale and personnel: a single administrator lectures to approximately 1000 students, and no teachers held the required qualifications. They were concerned that these constraints would hinder the program's goals. Participant 1 explained, "Just having one person for the entire grade or school probably means they can't cover all the content." Participant 2 suggested, "If a professional teacher were assigned, their research might be more in-depth, and they might also ... be suitable for more people."

Despite this lack of career-specific expertise, students highly valued the teachers' efforts to bridge the gap with their own professional and life experiences. Participant 9 commented, "I felt they also made ample preparations ... the teachers first acquired knowledge related to career planning before teaching it to us." Participant 8 similarly appreciated the guest speakers: "The guest speakers were also quite responsible. They spoke with a lot of energy and taught very attentively."

Six students mentioned that, besides course handouts, the school provided almost no other *learning materials*. They expressed a strong need for additional resources, noting that they were encountering career planning for the first time. Participant 1 recalled, "There was a document related to universities. If anyone needed it, they could ask him (the teacher) for it... But there were no other materials." Participant 11 emphasized, "We didn't have any at the time . .. . I feel that these kinds of resources are quite needed."

Overall, participants perceived the program as constrained by limited time, staffing, and learning resources, which restricted its depth and effectiveness despite recognized value.

3.3 | Process

3.3.1 | Program Delivery

Participants reported that the program's delivery focused on transmitting information for decision-making and was predominantly lecture-based. Although it occasionally incorporated self-exploration assessments and guest speakers, it provided minimal personalized guidance. While understanding the school's resource constraints and grateful for the help, participants ultimately felt that this standardized approach was insufficient.

The program covered *targeted content* that connected high school subjects, representative college majors, future vocational life, and personal traits. Eleven participants found this relevant for decision-making. Participant 8 considered the program "taught a lot of things I didn't know, such as college life and what college classes are like… as well as the pros and cons of various majors or job options after graduation." Participant 9 emphasized its value in exploring "interests, abilities, and values, and how to align these with career choices."

However, six participants pointed out that the content lacked depth and connection to the real world of work. As Participant 12 reflected, "What it taught was quite surface-level and very general. It was well-suited to that time in high school when we had no prior exposure and were just starting to plan." Participant 13 added, "They didn't explain what the job actually involves, so it just felt really abstract and vague."

Participants perceived that the program primarily used *direct instruction*. As Participant 15 noted: "You just listen to whatever the teacher teaches." Only Participant 8 considered this acceptable given the limited time and staff, stating, "With so many students in the school, the only practical option is to gather everyone together for a lecture." Occasionally, exploratory activities were incorporated, leaving a deeper impression. Participant 5 recalled, "The teaching method was rather monotonous, but under the teacher's guidance, we still played some small games."

Still, 11 participants believed that the program should adopt specialized methods to improve teaching effectiveness. Participant 2 highlighted the inefficiency of the lecture under intense academic pressure: "Because his method was face-to-face lecturing, but high school studies are too intense, listening to him talk might waste some time." Interestingly, when evaluating retrospectively, some participants used terminology from their college career program to articulate their high school challenges. As Participant 9 illustrated: "Career planning education needs to include self-positioning and assessment. This requires students to engage in self-reflection and exploration, which is quite difficult for them."

Impersonal support during the learning process was a major concern for ten participants. They understood that it was difficult for teachers to provide individualized guidance to large classes. As Participant 11 noted, "Because there were about 70 students in my class at the time, it was actually very difficult for the homeroom teacher to understand everyone's thoughts." However, they still strongly desired support to ease their anxiety about crucial decisions. Participant 12 complained, "There was no planning based on students' specific situations… they could only give broad advice to the majority."

Only Participant 3 engaged in an in-depth discussion with her class teacher regarding specific career pathways. Recalling how her teacher outlined the possibilities of becoming a dance teacher versus a professional dancer, she stated, "I also reflected on what the teacher told me," noting that this personalized guidance met her needs to a certain degree.

In short, although the program succeeded in providing a foundational overview of career planning, its standardized delivery was not structurally designed for individualized supports.

3.3.2 | Student Engagement

The program successfully fostered varying levels of *emotional engagement* among 12 participants. Ten participants appreciated its core objective of facilitating academic decision-making, which provided crucial support during a high-stress period. Participant 2 emphasized, "I was definitely feeling lost at the time. It felt a bit like a beam of life-saving light shining down." Participant 14 added, "It is a very major matter. Having the teacher talk about these things at that time makes you feel a bit more grounded." In contrast, Participants 1 and 4 engaged with the program outside its intended objectives. They viewed it as a temporary reprieve from exam pressure, with Participant 4 noting, "I quite liked participating, because it was a chance to relax a little bit."

Three participants (P6, 7, and 10) expressed growing complaints as the program progressed. As Participant 6 admitted, "I'm not entirely satisfied with the content either," whereas Participant 7 criticized it as "very superficial," and "not achieving the goal it had envisioned." This critique extended to the instructional design, with Participant 10 bluntly stating that the teaching method was "not right" and "boring."

All participants demonstrated appropriate behavioral engagement at the outset due to their recognition of the program's necessity and limited prior exposure. Participant 3 recalled, "I think I listened to the program the most attentively at the time," and Participant 5 noted, "I was willing to participate because it felt quite novel."

However, as the content lacked detail and the delivery methods lacked stimulation, engagement gradually declined. Participant 3 admitted, "I found it a bit dull and got distracted after a while." Participant 6 experienced a similar shift: "I was willing to participate at the time … but later, I realized the content wasn't particularly detailed, and I stopped paying much attention."

Regarding *cognitive engagement*, participants exhibited varying levels of agency in extending program learning into daily life. Seven participants demonstrated self-regulation in career planning. Participant 1 noted, "It still needed us to figure things out on our own… Through their (classmates') sharing and a deeper self-understanding, I gradually realized that I also needed a plan like this to guide me." Participant 9 explained that the program "gave us an initial idea about the future, so we would also think more about it during our daily studies." Others maintained a critical perspective. As Participant 15 stated, "It's pretty fun, but I don't think we should rely on those things entirely."

Conversely, some participants demonstrated relatively passive cognitive engagement, attributing this to the program's delivery or to examination preparation pressure. Participant 4 mentioned he almost gave up a preferred major because the teacher said, "If you can't learn it well, you won't have much to do besides sales." Participant 5 attributed her insufficient understanding of actual jobs to the program content being "too general" and without "any specific job roles covered." For Participant 7, academic pressure simply overshadowed career planning: "After choosing my examination subjects, I rarely thought about this matter anymore …. I believed that I should just focus on studying."

In essence, the program's lack of depth and personalization diminished initial emotional and behavioral engagement, leaving cognitive engagement largely dependent on students' own agency.

3.4 | Product

Participants acknowledged that their decision-making was facilitated by the career education program but constrained by the college admissions system and family influences.

3.4.1 | Empowerment

Fourteen participants specified how the program empowered them in navigating the complex choices of their final high school year. Four participants believed that the program provided necessary information and *expanded options* when selecting examination subjects and college majors. Participant 1 noted, "I might have originally known only ten paths, but now it has shown me there are more paths I can take." Participant 12 emphasized the importance of information for autonomous decision-making: "You can only make choices based on your understanding of certain majors. If you don't understand them, there is no way to choose."

Six students reported gaining clearer s*elf-understanding*, which they sought to integrate into their current and subsequent career planning. As Participant 5 elaborated:

> I might have had relatively few ideas during the program. When actually facing a choice, I would recall the teacher said in class that I could choose what suited me better based on personality assessments.

Participant 11 also noted that their understanding of "the connection between my own interests and the majors I would study in the future" could be sustained.

Eight participants considered their selection of examination subjects and college major as *informed decision-making* supported by the program, driven by a heightened awareness of autonomy in decision-making. As Participant 5 stated: "Before taking this course, I unconsciously thought that this was something to consider in college …. But that's not the case." Participants' independence was enhanced. Participant 7 explained:

> I only wanted to choose the science track before, because everyone said it was easy to find a job and the salary was high. But in this program, I gained a deeper, more comprehensive understanding and felt that I shouldn't just follow the crowd.

Four participants (P1, P8, P12, and P14) reported that the program provided a *sense of direction* during the stressful examination preparation process, showing them that their current hard work had a clear purpose. Participant 1 noted, "It resolved my dilemma of having no motivation in my senior year and not knowing which direction to pursue. Through this program, I obtained a goal." Participant 14 added, "My entire high school period was relatively confusing. This course did help to some extent with … my next steps!"

Ultimately, the program empowered students to navigate their futures by equipping them with broader options, clearer self-insight, an informed decision-making framework, and an enhanced sense of direction.

3.4.2 | Limited Adaptability

Although recognizing certain benefits of the program, six participants described significant difficulties in navigating the complex realities of the college admission system and family involvement. Despite the career-planning principles they learned, these participants found their decision-making constrained by systemic and familial factors, demonstrating the program's limited adaptability in high-stakes contexts.

Participants 2, 10, and 15 noted that *systemic constraints* ultimately overrode their emerging career plans, leading them to prioritize Gaokao scores over their interests. Participants 2 and 10 did not achieve their ideal exam scores. Participant 2 compromised on her preferred major to secure university admission: "I figured I might give it up and choose a different major." Participant 10 left the decision to her family, perceiving herself as uncompetitive for any targeted majors: "I thought about just picking a major at random at the time." Participant 15 chose exam subjects based on popularity rather than the self-knowledge gained from the program, ultimately leading to a major mismatch: "I chose the major myself, but I think I made a wrong decision."

These participants felt stuck by their current college majors. Participant 2 expressed a loss of direction: "Well… I want to study, but I can't focus. I'm not sure where to begin." Participant 15 highlighted the ongoing struggle to adjust: "I still want to stabilize myself a bit, but I feel the choices in between are just too difficult," indicating a need for further support to navigate these post-decision dilemmas.

Furthermore, Participants 7, 8, 10, and 12 highlighted *family involvement* in their decision-making, which often eclipsed the program's impact. Similar to Participant 10, Participant 7 avoided choosing a college major until his exam scores were released, ultimately deferring to his sister's opinion. Participant 12 considered that although the program was helpful, "ultimately, the decision on which major to choose should be based on our family." Even Participant 8, who benefited from supportive parents, observed how prevalent parental override was for others: "They completely respect my thoughts … but others' parents … may know which majors or industries are easier to make money, and then force their children to choose those majors."

In short, these experiences reveal a critical gap between intended program outcomes and entrenched admission systems and family dynamics, leaving students unequipped to navigate the complex compromises of their actual academic trajectories.

4 | Discussion

The current study examined graduates' perceptions in Mainland China regarding a typical career education program in a high-stakes academic high school. Findings indicated that the program functioned as a vital resource for students' decision-making and personal growth. The reformed examination system provided a structural impetus for implementing career education while remaining a significant constraint on both students' engagement and schools' practices. These findings underscored the need to balance meeting immediate examination requirements with fostering long-term career development in high-stakes academic contexts.

4.1 | Washback Effects on School Practices

The reformed high-stakes examination generated complex washback effects by imposing dual accountability on schools: delivering excellent examination results (Chiang 2018; Luo and Cao 2024; Meng et al. 2021) while simultaneously supporting students' holistic development through crucial academic decision-making (Bersan et al. 2024; Dodd and Hooley 2018; Wong et al. 2023). In response, the case school subordinated career education to examination success, as evident in the limited resources allocated to the program and the alignment of its objectives with improving examination outcomes. As recent studies have noted, the school's adoption of large lectures and standardized guidance is common in Mainland China, exemplifying efforts to meet accountability goals under resource constraints (Pang 2019; Wang and Yang 2023).

Furthermore, teachers demonstrated both devotion and goal-oriented pedagogy in program delivery. Previous research indicated that such goal-oriented pedagogy is widely adopted among high school teachers in systems prioritizing efficiency and measurable outcomes (Chiang 2018; Meng et al. 2021). The current study found that exam-oriented practices permeated career education. Although this approach ensured effectiveness in providing information and skills, it simultaneously reinforced a teacher-centered dynamic (Pang 2019; Wang and Yang 2023) at the expense of fostering students' learning engagement.

4.2 | Washback on Students' Expectations and Engagement

Findings indicate that the reformed examination system significantly raised students' attention to career planning. However, learning and applying career planning proved challenging. Previous research indicates that students' awareness and skills for autonomous personal development remain underdeveloped, partially due to the school system's prioritization of rote academic achievement and exam performance (Sun 2023; Cheng and Hamid 2025). Consistently, this study found that no participants had been exposed to career planning before high school. However, they were required to make critical decisions about exam subjects, which directly shaped their college majors and future careers, within the first year.

Although gradually aware of the complexity of academic decision-making, only a few students demonstrated self-regulation, in contrast to their major demands for detailed, practical vocational information. This specific engagement pattern corroborates prior research showing that students are forced toward outcome-driven tasks under time constraints and intense competition during examination preparation (Wang et al. 2022; Yu and Su 2024). It also echoes findings that introducing complex career choices within tight timeframes leaves students overwhelmed (Gu et al. 2020).

4.3 | Outcomes and Limitations of the Career Education Program

This study found that the case school's career program fulfilled its intended objectives by addressing examination requirements and fostering students' personal growth. Specifically, the program empowered students to make crucial decisions, though constrained by examination pressure and its delivery method. This finding aligns with previous research showing that brief career education programs can provide students with a fundamental understanding of the decision-making process in high-stakes academic contexts (Gu et al. 2020).

Additionally, the program helped students find meaning during this stressful period. Previous studies indicated that high-stakes examination pressure may hinder intrinsic motivation, learning autonomy, and mental health (Fu 2024; Heger 2017). In contrast, students perceived the career program as a designated space cultivating a sense of meaning by allowing them to reflect on their futures and exercise autonomy.

However, synthesizing the findings through the CIPP lens revealed a boundary between the program's design and systemic realities. First, constrained by input resources and a large-lecture format, the program was structurally positioned to offer foundational guidance rather than in-depth career exploration. Although previous research highlights the value of participatory learning in career education (Silipo and Caldon-Ruggles 2021;

Yoel and Dori 2022), the case school's design prioritized broad coverage.

Furthermore, the general content lacked components to address examination under-performance or family involvement. Previous research has shown that the complexity of students' career planning is heightened when future trajectories are predominantly determined by a single exam score (Cheng and Hamid 2025). Although concepts like career adaptability (Savickas 2005) are widely discussed in China (Chen et al. 2020), coping strategies for these systemic pressures have not been sufficiently integrated into school practices (Wang and Yang 2023). This reveals a significant gap between the program's intended introductory outcomes and the complex, high-stakes decision-making ecosystem, suggesting the need for interdisciplinary cooperation to extend current practices (Zhu et al. 2022; Wang and Yang 2023).

4.4 | Practical Implications

To address the specific challenges inherent in balancing schools' accountability for examination outcomes with students' holistic development, this study proposes several implications for designing and delivering career education in high-stakes contexts.

First, schools should adopt phased career education aligned with students' evolving needs while accounting for the pressure of exam preparation and subject selection (Gu et al. 2020). Educational systems worldwide, such as faculty-specific admissions in East Asia, A-Level and IB subject choices, or course planning for competitive US college majors, require both high-stakes examination performance and early academic decisions that limit future options. Therefore, school-based professionals must integrate career education with students' high-stakes decisions and examination timelines. Although the early program phases should focus on self-awareness and exploration, later phases can emphasize decision-making skills. This gradual approach prepares students without overwhelming them during the peak of exam preparation. Furthermore, given the significant uncertainty in the college admission, schools should foster students' resilience and career adaptability (Savickas 2005) to help them navigate potential setbacks, such as not being admitted to their preferred major.

Second, school-based career education should include structured components for parent engagement to create a supportive ecosystem for students' career development. Workshops or information sessions can help parents understand their children's career development process and learn how to provide constructive support. These components can foster collaborative parent–child interactions regarding career choices.

Third, investment in teachers' professional development is crucial to maximize program effectiveness within limited instructional hours. Teachers providing career education must receive targeted training that expands their pedagogical repertoires beyond subject-matter expertise. Specifically, this training should empower teachers to shift from mere information-dispensing to constructivist approaches (Healy 2023; Kim and Lee 2023; Pang 2019). Teachers should develop practical techniques for experiential learning, including facilitating personal reflection, guiding peer discussions, and conducting scenario-based job analyses grounded in adolescent career development theories (Chang and Lee 2023).

Finally, the effective delivery requires the active participation of dedicated career professionals (e.g., career counselors and external consultants). Students' dual burden of selecting a major before adequate career exposure and securing scores for competitive college admission is severely amplified in contexts like Mainland China. However, even in systems like the United States, the intense effort required for high standardized test scores (e.g., SAT or ACT) similarly crowds out time for meaningful career exploration. Moreover, the scarcity of competitive college admissions and scholarships may force students to prioritize competition over pursuing a suitable major, which is the most critical developmental task of this life stage. To mitigate this, career professionals should provide strategic guidance to students and their families to balance academic survival with identity exploration as an essential part of school-based career education.

4.5 | Limitations and Future Directions

This study has several limitations. First, it relies solely on interview data to provide an in-depth investigation of students' perceptions of their learning experiences. Although qualitative interviews are well-suited for capturing in-depth individual reflections, incorporating methodological triangulation could further enhance the trustworthiness of the findings. Future research should consider integrating additional data sources, such as surveys and academic records, to corroborate these self-reported experiences.

Second, this study adopts a retrospective design, inviting graduates to reflect on their high school experiences. Such recollections may have evolved over time and been influenced by subsequent university experiences or current life stages, potentially diverging from their original perceptions. Future studies may adopt a longitudinal research design to track students from high school into higher education, capturing this evolving process and providing further insights into long-term program impacts.

Finally, this study focused on a single well-established high school in Eastern Mainland China. Given regional disparities in educational resources and policy implementation, future studies should include schools across diverse geographical and socioeconomic contexts to better understand the varied challenges of implementing career education within a high-stakes examination system.

5 | Conclusion

In conclusion, drawing on graduates' reflections, this study reveals the complex dynamics of implementing career education within high-stakes examination contexts. The findings underscore that despite systemic constraints, the school-based career education program remains critical to supporting students' personal development and career navigation during highly stressful periods. These findings suggest that enhancing the effectiveness of career education programs requires innovations in program planning, targeted professional development for teachers, and the intentional engagement of career professionals. Future research should continue to explore the unique challenges and effective factors of career programs across global educational systems that demand both high-stakes examination performance and early academic decision-making.

Funding

The authors have nothing to report.

Ethics Statement

Ethical approval for this study was obtained from the Institutional Review Board of City University of Macau before data collection.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data that support the findings of the study are available from the corresponding author upon reasonable request.

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